

Relaxation and Odd-Even Effect in Fermi Liquids

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Outline

1

Introduction

- k -space
- Fermi Dirac distribution
- Boltzmann equation

2

Recovering equations

- Linearization of the collision integral
- Eigenvalue problem

3

Odd-Even effect

- Zero modes
- Kinematic Constraints

1

Introduction

1.1

***k*-space**

k -space

- Analogous to momentum space
- Each point $\mathbf{k}$ in k -space corresponds to a wave $e^{i\mathbf{k}}$
- **Fourier transform** converts between k -space to real space

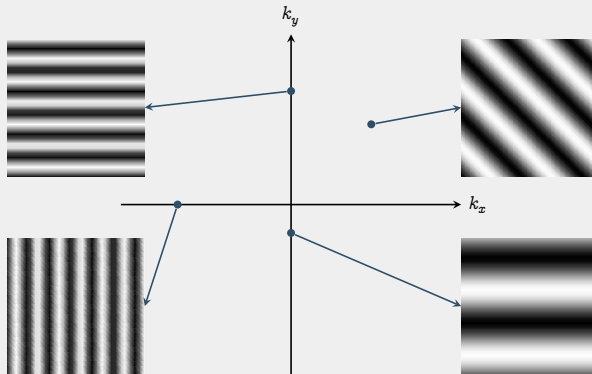


Figure 1: REPRESENTATIVE WAVES OF k -SPACE

- Distance from origin $\propto$ Wave frequency
- Direction on k -space $\implies$ Wave direction

1.2

Fermi Dirac distribution

Fermi-Dirac distribution

$$f_0(\mathbf{p}) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}) - \mu)] + 1}$$

1

- $\beta = 1/k_B T$ is the inverse temperature
- $\varepsilon(\mathbf{p}) = \frac{\hbar^2 \mathbf{p}^2}{2m}$ is the energy of particles with momentum $\mathbf{p}$
- μ is the chemical potential

The Fermi-Dirac distribution is in **equilibrium**.

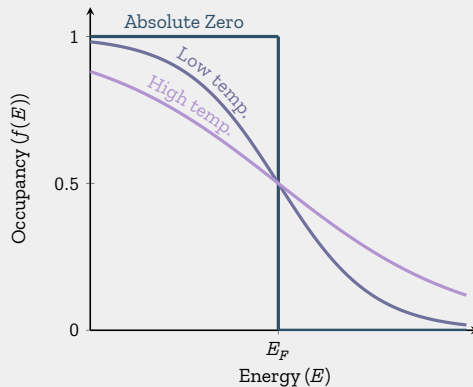


Figure 2: FERMI-DIRAC DISTRIBUTION AT DIFFERENT TEMPERATURES

- At $T = 0$, all states below the Fermi energy are filled, all states above are empty.
- As T increases, the distribution smooths out.

The Fermi surface is the set of points in k -space where the energy of the electrons is equal to the Fermi energy.

Under no thermal gradients and external force, the Fermi surface is a a circle in 2D k -space.

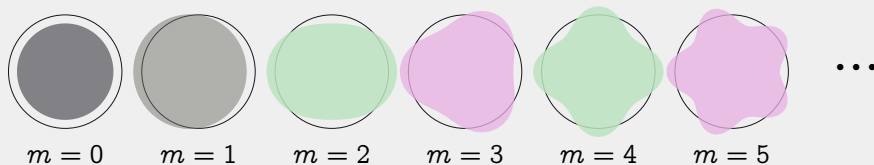


Figure 3: CIRCULAR HARMONICAL PERTURBATIONS OF THE FERMI SURFACE

The Fermi surface may be perturbed by external fields. Since a perturbation has to be smooth, we can expand the perturbation in terms of circular harmonics $e^{im\theta}$ in 2D.

1.3

Boltzmann equation

The Boltzmann equation describes the time evolution of the distribution function $f(t, \mathbf{p})$ in phase space:

$$\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + \mathbf{F} \cdot \frac{\partial}{\partial \hbar \mathbf{p}} \right) f(t, \mathbf{x}, \mathbf{p}) = \hat{\mathcal{J}}[f(t, \mathbf{x}, \mathbf{p})] \quad 2$$

- $\frac{\partial f}{\partial t}$ is the time evolution of the distribution function
- $\mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{r}}$ is the advection term (by, for example, thermal gradients)
- $\mathbf{F} \cdot \frac{\partial f}{\partial \hbar \mathbf{p}}$ is the external force term
- $\hat{\mathcal{J}}[f]$ is the collision integral, which describes the effect of collisions on the distribution function

2

Recovering equations

2.1

Linearization of the collision integral

Ideas

- Consider **only perturbation near Fermi's equilibrium**

$$f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$$

- Transform the integrals into a linear operator

Preparation

Fermi equilibrium:
$$f_0(\mathbf{p}) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}) - \mu)] + 1}$$

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{1}{T} f_0(1 - f_0) \quad 3$$

$$\frac{\partial f_0}{\partial \mu} = \frac{1}{T} f_0(1 - f_0) \quad 4$$

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{\partial f_0}{\partial \mu} \quad 5$$

$$f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$$

Perturbation

6

$$\approx f_0(\mathbf{p}) + \frac{\partial f_0}{\partial \mu} \delta \mu$$

Taylor series. Fix ε

7

$$\therefore \delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} \delta \mu$$

8

$$f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$$

Perturbation

6

$$\approx f_0(\mathbf{p}) + \frac{\partial f_0}{\partial \mu} \delta \mu$$

Taylor series. Fix ε

7

$$\therefore \delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} \delta \mu$$

8

Perturbation of interest: $\delta \mu = T \psi$ (temperature dependent change in chemical potential)

$$\delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} T \psi \quad 9$$

$$= \left(-\frac{1}{T} f_0 (1 - f_0) \right) \times T \psi \quad 10$$

$$= f_0 (1 - f_0) \psi \quad 11$$

Collision integral

$$\hat{\mathcal{J}}[f] = - \int \dots \int \frac{1}{(2\pi)^6} d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) \mathcal{B}, \quad 12$$

$$\mathcal{B} = f_1 f_2 (1 - f_{1'}) (1 - f_{2'}) - f_{1'} f_{2'} (1 - f_1) (1 - f_2), \quad 13$$

$$f_i = f(\mathbf{p}_i), \quad f_{0,i} = f_0(\mathbf{p}_i). \quad 14$$

Linearization

Taylor expansion of $\mathcal{B}$ at f_0 :

$$\mathcal{B} = \mathcal{B}_{f_0} + \mathcal{B}_{\text{lin}} + \mathcal{B}_{\text{non-lin}} \quad 15$$

The linear part is:

$$\mathcal{B}_{\text{lin}} = \left(\frac{\partial \mathcal{B}}{\partial f_1} \right)_{f_0} \delta f_1 + \left(\frac{\partial \mathcal{B}}{\partial f_2} \right)_{f_0} \delta f_2 + \left(\frac{\partial \mathcal{B}}{\partial f'_1} \right)_{f_0} \delta f_{1'} + \left(\frac{\partial \mathcal{B}}{\partial f'_2} \right)_{f_0} \delta f_{2'} \quad 16$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} = f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1'}f_{0,2'}(1 - f_{0,2}), \quad 17$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} = f_{0,1}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1}f_{0,2}(1 - f_{0,1}), \quad 18$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_{1'}}\right)_{f_0} = -f_{0,1}f_{0,2}(1 - f_{0,2'}) - f_{0,2'}(1 - f_{0,1})(1 - f_{0,2}), \quad 19$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_{2'}}\right)_{f_0} = -f_{0,1}f_{0,2}(1 - f_{0,1'}) - f_{0,1'}(1 - f_{0,1})(1 - f_{0,2}). \quad 20$$

Substitute $\delta f_i = f_{0,i}(1 - f_{0,i})\psi_i$. E.g.,

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 = \left[f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1'}f_{0,2'}(1 - f_{0,2}) \right] \times f_{0,1}(1 - f_{0,1})\psi_1 \quad 21$$

$$= \left[f_{0,1}f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'})(1 - f_{0,1}) + f_{0,1}f_{0,1'}f_{0,2'}(1 - f_{0,1})(1 - f_{0,2}) \right] \psi_1$$

$$= \left(f_{0,1}f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) \right) \left[(1 - f_{0,1}) + f_{0,1} \right] \psi_1 \quad 22$$

$$= \left(f_{0,1}f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) \right) \psi_1 \quad 23$$

Define $F_{121'2'} = f_{0,1}f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) = f_{0,1'}f_{0,2'}(1 - f_{0,1})(1 - f_{0,2})$. So that

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 = F_{121'2'}\psi_1 \quad 24$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_i}\right)_{f_0} \delta f_i = F_{121'2'} \psi_i \quad 25$$

Hence,

$$\mathcal{B}_{\text{lin}} = \left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 + \left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} \delta f_2 + \left(\frac{\partial \mathcal{B}}{\partial f_{1'}}\right)_{f_0} \delta f_{1'} + \left(\frac{\partial \mathcal{B}}{\partial f_{2'}}\right)_{f_0} \delta f_{2'} \quad 26$$

$$= F_{121'2'} [\psi_1 + \psi_2 - \psi_{1'} - \psi_{2'}] \quad 27$$

$$= F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}'_1) - \psi(\mathbf{p}'_2)] \quad 28$$

$$\hat{\mathcal{J}}[f] = -\frac{1}{(2\pi)^6} \int \dots \int d\mathbf{p}_2 d\mathbf{p}_1 d\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) \mathcal{B} \quad 29$$

$$= -\frac{1}{(2\pi)^6} \int \dots \int d\mathbf{p}_2 d\mathbf{p}_1 d\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) (\mathcal{B}_{f_0} + \mathcal{B}_{\text{lin}} + \mathcal{B}_{\text{non-lin}}) \quad 30$$

- Collision integral vanishes at equilibrium: $\hat{\mathcal{J}}[f_0] = 0$
- Throw out $\mathcal{B}_{\text{non-lin}}$

$$\begin{aligned} \hat{\mathcal{J}}[f] = & -\frac{1}{(2\pi)^6} \int \dots \int d\mathbf{p}_2 d\mathbf{p}_1 d\mathbf{p}'_2 \\ & \times W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}'_1) - \psi(\mathbf{p}'_2)] \quad 31 \end{aligned}$$

Expression of $F_{121'2'}$

- For convenience, write $F_{121'2'}$ as **total momentum** (P) and **center of momentum** (Q)

$$\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2, \quad \mathbf{Q} = \frac{1}{2}\mathbf{p}_1 - \frac{1}{2}\mathbf{p}_2.$$

32

$$\mathbf{p}_1 = \frac{1}{2}\mathbf{P} - \mathbf{Q}, \quad 33$$

$$\mathbf{p}_2 = \frac{1}{2}\mathbf{P} + \mathbf{Q}, \quad 34$$

$$\mathbf{P} \cdot \mathbf{Q} = \frac{1}{2}(p_1^2 - p_2^2). \quad 35$$

$$\varepsilon(\mathbf{p}_1) = \frac{\hbar^2 p_1^2}{2m^*} = \frac{\hbar^2}{2m^*} \left(\frac{1}{4}P^2 + Q^2 - \mathbf{P} \cdot \mathbf{Q} \right), \quad 36$$

$$\varepsilon(\mathbf{p}_2) = \frac{\hbar^2 p_2^2}{2m^*} = \frac{\hbar^2}{2m^*} \left(\frac{1}{4}P^2 + Q^2 + \mathbf{P} \cdot \mathbf{Q} \right). \quad 37$$

Define¹

$$\xi = \frac{\hbar^2 \beta}{2m^*}, \quad \text{and} \quad X = \beta \left(\frac{\hbar^2 P^2}{8m^*} + \frac{\hbar^2 Q^2}{2m^*} - \mu \right) \quad 38$$

Then,

$$\beta(\varepsilon(\mathbf{p}_1) - \mu) = \beta \left(\frac{\hbar^2}{8m^*} P^2 + \frac{\hbar^2}{2m^*} Q^2 - \mu \right) - \frac{\hbar^2 \beta}{2m^*} \mathbf{P} \cdot \mathbf{Q} \quad 39$$

$$= X - \xi \mathbf{P} \cdot \mathbf{Q}. \quad 40$$

Similarly,

$$\beta(\varepsilon(\mathbf{p}_2) - \mu) = X + \xi \mathbf{P} \cdot \mathbf{Q}. \quad 41$$

¹ X can possibly be interpreted as: *average deviation from background chemical potential*

Hence,

$$f_0(\mathbf{p}_1) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}_1) - \mu)]} = \frac{1}{\exp[X - \xi \mathbf{P} \cdot \mathbf{Q}]}.$$
 42

$$f_0(\mathbf{p}_2) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}_1) - \mu)]} = \frac{1}{\exp[X + \xi \mathbf{P} \cdot \mathbf{Q}]}$$
 43

Recall that $F_{121'2'} = f_0(\mathbf{p}_1)f_0(\mathbf{p}_2)(1 - f_0(\mathbf{p}'_1))(1 - f_0(\mathbf{p}'_2))$

$$f_0(\mathbf{p}_1)f_0(\mathbf{p}_2) = \frac{1}{\exp[X - \xi \mathbf{P} \cdot \mathbf{Q}] \exp[X + \xi \mathbf{P} \cdot \mathbf{Q}]} \quad 44$$

$$= \frac{1}{\exp[2X] + \exp[X](\exp[\xi \mathbf{P} \cdot \mathbf{Q}] + \exp[-\xi \mathbf{P} \cdot \mathbf{Q}]) + 1} \quad 45$$

$$= \frac{1}{\exp[2X] + 2 \exp[X] \cosh(\xi \mathbf{P} \cdot \mathbf{Q}) + 1} \quad 46$$

$$= \frac{1}{2(\cosh X + \cosh(\xi \mathbf{P} \cdot \mathbf{Q}))} \quad 47$$

By symmetry,

$$(1 - f_0(\mathbf{p}'_1))(1 - f_0(\mathbf{p}'_2)) = \frac{1}{2(\cosh X + \cosh(\xi \mathbf{P}' \cdot \mathbf{Q}'))} \quad 48$$

$$F_{121'2'} = \frac{1}{4[\cosh X + \cosh(\xi \mathbf{P} \cdot \mathbf{Q})][\cosh X + \cosh(\xi \mathbf{P}' \cdot \mathbf{Q}')] } \quad 49$$

Expression of W - the collision rate

$$W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) = \frac{2\pi}{\hbar} |V|^2 \times (2\pi)^2 \\ \times \delta(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}'_1 - \mathbf{p}'_2) \delta(\varepsilon(\mathbf{p}_1) + \varepsilon(\mathbf{p}_2) - \varepsilon(\mathbf{p}'_1) - \varepsilon(\mathbf{p}'_2)) \quad 50$$

$$\hat{\mathcal{J}}[f(t, \mathbf{p}_1)] = - \frac{1}{(2\pi)^6} \int \dots \int d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 \times \frac{2\pi}{\hbar} |V|^2 \times (2\pi)^2 \\ \times \delta(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}'_1 - \mathbf{p}'_2) \delta(\varepsilon(\mathbf{p}_1) + \varepsilon(\mathbf{p}_2) - \varepsilon(\mathbf{p}'_1) - \varepsilon(\mathbf{p}'_2)) \\ \times F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}'_1) - \psi(\mathbf{p}'_2)] \quad 51$$

Change of coordinates: $\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2$, $\mathbf{Q} = \frac{1}{2}(\mathbf{p}_1 - \mathbf{p}_2)$.

$$\begin{aligned}
 \hat{\mathcal{J}}[f(t, \mathbf{p}_1)] &= -\frac{1}{(2\pi)^3 \hbar} \int \dots \int d\mathbf{p}_2 d\mathbf{P}' d\mathbf{Q}' \delta(\mathbf{P} - \mathbf{P}') \delta\left(\frac{\hbar^2}{2m^*}(p_1^2 + p_2^2 - p_1'^2 - p_2'^2)\right) \\
 &\quad \times |V|^2 F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')] \\
 &= -\frac{1}{(2\pi)^3 \hbar} \int \dots \int d\mathbf{p}_2 d\mathbf{Q}' \delta\left(\frac{\hbar^2}{4m^*}(4Q^2 - 4Q'^2)\right) \\
 &\quad \times |V|^2 F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')] \\
 &= -\frac{1}{(2\pi)^3 \hbar} \int \dots \int d\mathbf{p}_2 d\mathbf{Q}' d\Omega \frac{m^*}{\hbar^2} \delta(Q^2 - Q'^2) \\
 &\quad \times |V|^2 F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')] \\
 &= -\frac{m^*}{4\pi \hbar^3} \int \dots \int \frac{d\mathbf{p}_2}{(2\pi)^2} \int d\Omega |V|^2 F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')]
 \end{aligned}$$

2.2

Eigenvalue problem

Reducing the Boltzmann equation

$$\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + \mathbf{F} \cdot \frac{\partial}{\partial \hbar \mathbf{p}} \right) f(t, \mathbf{r}, \mathbf{p}) = \hat{\mathcal{J}}[f(t, \mathbf{r}, \mathbf{p})] \quad 52$$

- No external force: $\mathbf{F} = 0$,
- No thermal gradient: $\frac{\partial f}{\partial \mathbf{r}} = 0$.

The equation reduces to

$$\frac{\partial f}{\partial t} = \hat{\mathcal{J}}[f] \quad 53$$

Recall that

- $f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$
- $\delta f(t, \mathbf{p}) = f_0(1 - f_0)\psi(t, \mathbf{p})$
- $\frac{\partial f_0}{\partial t} = 0, \hat{\mathcal{J}}[f_0] = 0$

Thus, the equation further reduces to

$$\frac{\partial \psi(t, \mathbf{p})}{\partial t} = \hat{\mathcal{J}}[\psi(t, \mathbf{p})]$$

Proposed ansatz $\psi(t, \mathbf{p}) = e^{-\gamma t} \phi(\mathbf{p})$ (Chemical potential decays with time)

So, the equation reduces to an eigenvalue problem:

$$\hat{\mathcal{T}}[\phi(\mathbf{p})] = -\gamma \phi(\mathbf{p})$$

Under the constraints of continuity and smoothness, the eigenfunctions are sums of circular harmonics $e^{im\theta}$ in 2D. Picking a test mode m , we obtain

$$\hat{\mathcal{J}}[e^{im\theta}] = -\gamma_m e^{im\theta} \quad 56$$

Our goal is to study the eigenvalues γ_m as a function of m and temperature T .

3

Odd-Even effect

3.1

Zero modes

We will first consider the zero modes, which are when $\gamma_m = 0$, associated with quantities that are conserved.

- $\gamma_0 = 0$ has two eigenfunctions: $\phi_0 = 1$ and $\phi_0 = \varepsilon(\mathbf{p})$, corresponding to the conservation of particle number and energy, respectively.
- $\gamma_1 = 0$ has two eigenfunctions: $\phi_1 = e^{i\theta}$ and $\phi_1 = e^{-i\theta}$, corresponding to the conservation of momentum in the x and y directions, respectively.

3.2

Kinematic Constraints

Near the Fermi surface,

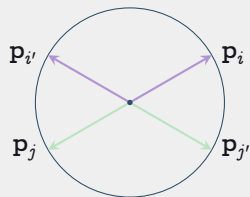
$$\varepsilon(\mathbf{p}_1) \approx \varepsilon(\mathbf{p}_2) \approx \varepsilon(\mathbf{p}'_1) \approx \varepsilon(\mathbf{p}'_2) \approx \varepsilon_F. \quad 57$$

Thus, **conservation of energy** implies **conservation of momentum magnitude**:

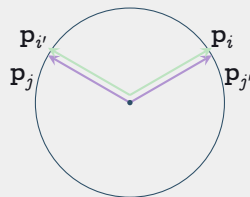
$$p_1 + p_2 = p'_1 + p'_2. \quad 58$$

Three types of collisions that satisfy this constraint:

- Head-on collisions: $\mathbf{p}_1 = -\mathbf{p}_2, \mathbf{p}'_1 = -\mathbf{p}'_2$
- Forward exchange: $\mathbf{p}'_1 = \mathbf{p}_2, \mathbf{p}'_2 = \mathbf{p}_1$
- No collision: $\mathbf{p}'_1 = \mathbf{p}_1, \mathbf{p}'_2 = \mathbf{p}_2$



Head-on process



Exchange process

Figure 4: RELEVANT COLLISION TYPES

No collisions: Has no contribution to the collision integral.

Forward exchange & No collisions: • $\mathbf{p}_1 = \mathbf{p}'_1, \mathbf{p}_2 = \mathbf{p}'_2$.

- $\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}'_1) - \psi(\mathbf{p}'_2) = 0^2$
- $\therefore$ These collisions has no contribution to the collision integral.

Only **head-on collision** contribute to the collision integral.

²Recall that this is a multiplier in the collision integral

Head on collision

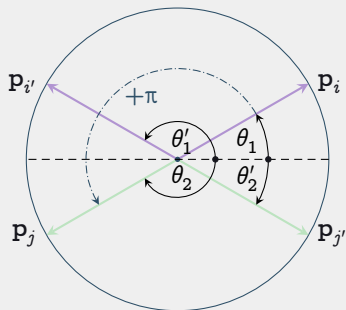


Figure 5: HEAD-ON COLLISION ANGLE

$$\theta_2 = \theta_1 + \pi, \quad \text{and} \quad \theta_2' = \theta_1' + \pi. \quad 59$$

Substituting test mode m , and recalling $e^{im(\theta+\pi)} = (-1)^m e^{im\theta}$, then

$$\begin{aligned} & \psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2') \\ &= (1 + (-1)^m) e^{im\theta_1} - (1 + (-1)^m) e^{im\theta_1'}. \end{aligned} \quad 60$$

m is odd: the term vanishes; thus the collision integral vanishes.

m is even: the term does not vanish.